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Surface active agents
**Determination of interfacial tension of solutions of surface
 active agents by the stirrup or ring method**
 English version of DIN EN 14210

Grenzflächenaktive Stoffe – Bestimmung der Grenzflächenspannung von grenzflächenaktiven Lösungen
 mittels Bügel- oder Ringverfahren

European Standard EN 14210 : 2003 has the status of a DIN Standard.

A comma is used as the decimal marker.

National foreword

This standard has been prepared by CEN/TC 276 'Surface active agents' (Secretariat: France).

The responsible German body involved in its preparation was the *Normenausschuss Materialprüfung* (Materials Testing Standards Committee), Technical Committee *Anwendungstechnische Prüfverfahren für Tenside*.

Amendments

DIN 53993, December 1997 edition, has been superseded by the specifications of EN 14210.

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English version

Surface active agents

**Determination of interfacial tension of solutions of
surface active agents by the stirrup or ring method**

Agents de surface – Détermination de
la tension interfaciale des solution
d'agents de surface par la méthode à
l'anneau ou l'étrier

Grenzflächenaktive Stoffe – Bestim-
mung der Grenzflächenspannung von
grenzflächenaktiven Lösungen mittels
Bügel- oder Ringverfahren

This European Standard was approved by CEN on 2003-10-01.

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CEN

European Committee for Standardization
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Europäisches Komitee für Normung

Management Centre: rue de Stassart 36, B-1050 Brussels

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Foreword

This document EN 14210:2003 has been prepared by Technical Committee CEN/TC 276 “Surface active agents”, the secretariat of which is held by AFNOR.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by June 2004, and conflicting national standards shall be withdrawn at the latest by June 2004.

This document includes a Bibliography.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard : Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Slovakia, Spain, Sweden, Switzerland and the United Kingdom.

1 Scope

This European Standard specifies test methods for the determination of interfacial tension in the range from 4 mN/m to 50 mN/m between two immiscible liquids which can also be free from surface active agents.

It is particularly suitable for determining the interfacial tension between water or aqueous solutions and organic liquids which are immiscible with water.

2 Principle

The maximum force, F , necessary to pull or to force a stirrup or ring of length, l , out of the interface between two liquids in the direction of the nonaqueous phase is measured. The interfacial tension, γ , is obtained by calculation where the following approximate equation (1) serves as base:

$$\gamma = \frac{F}{2l} \quad (1)$$

3 Apparatus

3.1 Tensiometer

3.1.1 General

The tensiometer shall be designed for use with a stirrup or ring and shall consist essentially of the parts specified in 3.1.2 and 3.1.3.

3.1.2 Dynamometer (e.g. torsion balance or inductive force transducer), capable for reading off the force to the nearest 0,1 mN/m over the effective length of the stirrup or ring.

3.1.3 Support for the measuring vessel, consisting of a horizontal platform which can be moved vertically up and down, e.g. with the aid of a precision drive.

3.2 Stirrup

The stirrup shall consist of platinum/iridium wire with a test wire diameter of 0,1 mm and a length of 20 mm to 40 mm. The length of the two branches shall be about 20 mm. Two small platinum balls can be attached to the lower ends of the branches as counterweights or, alternatively, the lower ends of the branches can be joined by a platinum wire (see Figure 1 as an example).

Dimensions in millimetres

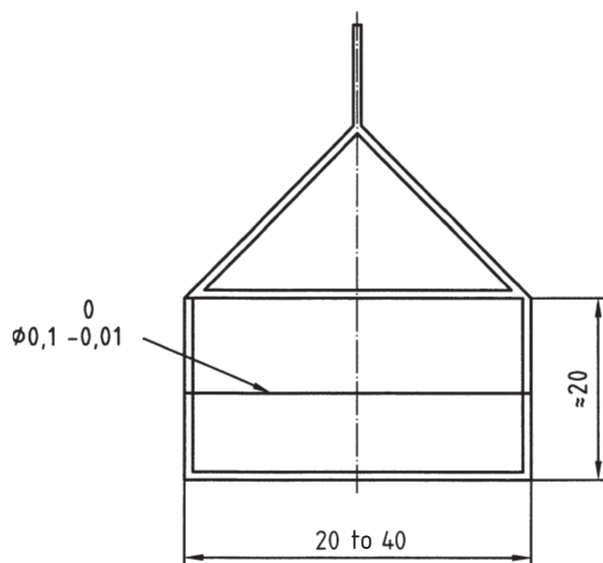


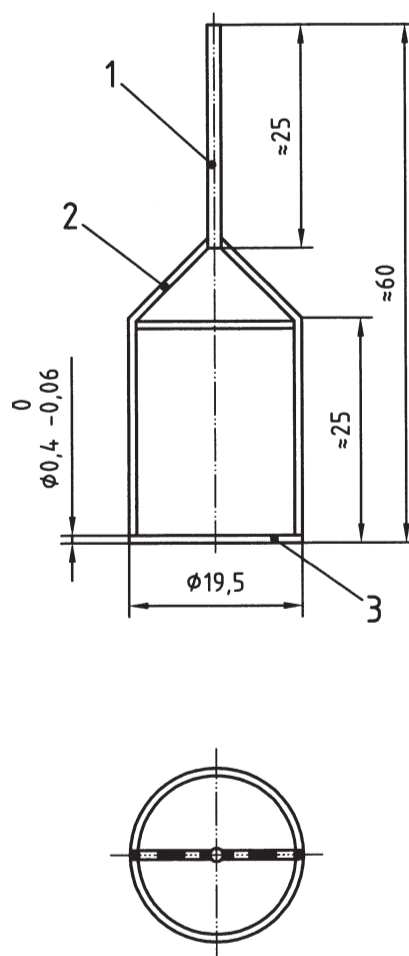
Figure 1 — Stirrup for measuring interfacial tension

NOTE The stirrup specified in EN 14370 for measuring the surface tension is identical with that one shown in Figure 1, except for the lengths of the measuring wire and the branches.

3.3 Ring

The ring shall consist of a platinum or platinum/iridium wire with a thickness less than or equal to 0,4 mm and a mean circumference of 60 mm. By means of a wire stirrup, it shall be attached in a horizontal position to a metal rod forming the link to the tensiometer (see Figure 2 as an example).

Dimensions in millimetres



Key

- 1 Metal rod
- 2 Wire loop
- 3 Ring

Figure 2 — Ring for measuring interfacial tension

NOTE The ring specified in EN 14370 for measuring the surface tension is identical with that one shown in Figure 2.

3.4 Sample vessel

The sample vessel for the liquid under test shall be a glass vessel which is suitable for use in a thermostat. It shall have a sufficient capacity for the liquid which, if necessary, shall be protected from evaporation. It shall be constructed such that the temperature of the gas phase above the liquid can also be kept constant by thermostat. There is no need to use a thermostat for measurement at room temperature. The sample vessel shall consist of a cylindrical glass container with an internal diameter of at least 45 mm for use with the ring method or stirrup method.

NOTE If a vessel of less than 45 mm diameter needs to be used (for example, if only small amounts of sample are available), wall effects can cause an error in the interfacial tension measurement. The magnitude of this error depends on the distance between the wall of the vessel and the edge of the ring or stirrup.

4 Preparation of apparatus

4.1 Cleaning

Clean the glass vessels carefully, first by thorough preliminary cleaning using a suitable solvent, for example propan-2-ol or propanone. Then rinse them repeatedly with freshly double distilled water and dry them. Clean and rinse the stirrups or rings in the same manner, and then heat them briefly to incandescence in a methanol or ethanol flame (the latter may be preferred since it has a low vapour pressure and higher flash point), preferably a non-luminous natural gas flame, ensuring they are only heated to red heat and never to white heat.

NOTE Should the stirrups or rings become contaminated with residues, for example from silicones or cationic surface active agents, it can be preferable to clean with a sulfo-chromic acid mixture or as specified in ISO 6889.

4.2 Measurement preparation

Double-distilled water or water of equivalent purity shall be used to prepare any aqueous solution required for the test.

The temperature of the liquids shall be kept constant to the nearest 0,5 °C during the measurement. The measurement temperature chosen shall be an appropriate one for the objective of a particular test. Measurements at room temperature present the fewest problems.

The interfacial tension changes with time and can be affected by the chemical nature of a surface active agent, its concentration, whether it is dissolved in the upper or lower phase, its adsorptive capacity and its adsorption rate during the change. So it is not possible to specify an appropriate time for starting the measurements. Therefore, several measurements of the interfacial tension as a function of time and, consequently, of the age of the interface shall be performed. After a time which is characteristic for every system, no further changes are observed in the interfacial tension since an equilibrium state is established. It is advantageous to use commercially available, automatic tensiometers to carry out such measurements.

Interfacial systems are extremely sensitive to contaminants. For this reason, no work involving volatile substances shall be carried out in the measurement room and the apparatus shall be protected by an enclosure of the type used for balances. This precaution also facilitates maintenance of a constant temperature.

5 Procedure

5.1 General

Two methods of measurement are described in 5.2 and 5.3. In the relative stirrup or ring method (see 5.2), the measured value is compared with the corresponding value determined on a system of two liquids with similar densities to those of the liquids under test densities and known interfacial tension. In the absolute ring method (5.3), account is taken of a factor for the apparatus determined using an adjustment weight and a Zuidema and Waters correction factor [1] when determining the interfacial tension. The Zuidema and Waters factor is based on the correction determined empirically by Harkins and Jordan [2] for measurements made by the ring method. Different measurement procedures are adopted, depending on whether the density of the aqueous phase is greater or less than that of the nonaqueous phase.

5.2 Relative stirrup or ring method

5.2.1 Method A

Method A is a method for systems in which the density of the aqueous phase is greater than that of the nonaqueous phase.

Use a pipette to transfer an amount of the aqueous phase about 15 mm deep to the measuring apparatus holding the tip of the pipette against the vessel wall, avoiding bubbles. Check that the test wire of the stirrup or ring is horizontal by using the surface of the liquid as a mirror when the stirrup or ring is just above it. Then slowly raise the platform and the measurement vessel until the stirrup test wire or ring is immersed in the aqueous phase. Then

add a roughly 15 mm deep layer of the nonaqueous phase, again avoiding bubbles, and leave for at least 5 min (see clause 3) to allow the liquids to become mutually saturated.

Then repeatedly lower the platform and raise it again until the film almost breaks. Read off the maximum value, W_2 , obtained on the tensiometer every time. Read off the value W_1 when the stirrup test wire or ring is immersed without a film in the upper phase to the same extent as when measuring W_2 . This gives the difference $\Delta W = W_2 - W_1$. If the same procedure is used for a system of two liquids with similar densities and known interfacial tension, γ_0 , (e.g. a value from the literature), ΔW_0 is obtained.

5.2.2 Method B

Method B is a method for systems in which the density of the aqueous phase is less than that of the nonaqueous phase.

Pour both the phases into the measurement vessel to a depth of 15 mm in each case, avoiding bubbles. After checking that the stirrup or ring is horizontal (as in 5.2.1), raise the platform and measuring vessel until the stirrup or ring is immersed in the upper phase and leave for at least 5 min (see clause 4) to allow the liquids to become mutually saturated. Then raise the platform further until the stirrup or ring passes through the interface and after that raise and lower it again until the film almost breaks. Read off the maximum value, W_2 , obtained in each case. Read off W_1 when the stirrup measuring wire or ring is immersed without a film in the lower phase to the same extent as in the measurement of W_2 . This yields the difference $\Delta W = W_2 - W_1$. Then determine ΔW_0 as in 5.2.1.

5.3 Absolute ring method

Calibrate the tensiometer by adding a weight with a known mass, m , of about 0,5 g to the ring and register the resultant reading, A_1 . Determine the maximum value, W_2 , for the formation of the film and the value, W_1 , without a film and take the difference, $\Delta W = W_2 - W_1$, as in the relative method (5.2.1 or 5.2.2).

6 Expression of results

6.1 General

In view of the mass of the stirrup or ring and of the fact that the magnitude of the forces encountered is low, it is advisable to express all the quantities in centimetres, grams, seconds (cgs units), e.g. $g = 981 \text{ cm/s}^2$, ρ in g/cm^3 and wire length and wire thickness in centimetres. This will give the interfacial tension in the SI unit millinewtons per meter ($1 \text{ mN/m} = 1 \text{ dyn/cm}$).

6.2 Relative stirrup or ring method

The required interfacial tension, γ , is given by equation (2) :

$$\gamma = \gamma_0 \frac{\Delta W}{\Delta W_0} \quad (2)$$

where

γ_0 is the value given in the literature for the interfacial tension of the reference system;

$\Delta W, \Delta W_0$ see 5.2.1 or 5.2.2.

6.3 Absolute ring method

Determine the factor, Φ , for the apparatus using equation (3):

$$\Phi = \frac{A_2}{A_1} \quad (3)$$

where

$$A_2 = \frac{m \cdot g}{2\pi D} \quad (4)$$

where

m is the mass of the additional mass;

g is the acceleration due to gravity;

D is the mean ring diameter;

A_1 is the reading on the tensiometer when the additional mass, m , is added (see 5.3).

The approximate value, γ , of the interfacial tension is then:

$$\gamma = \Delta W \cdot \Phi \quad (5)$$

To obtain the Zuidema and Waters [1] interfacial tension γ_{zw} , multiply the above value by the Zuidema and Waters correction factor, f , [1]:

$$f = 0,725 + \sqrt{\frac{4 \times 0,360 \, 7 \, \gamma}{D^2 g \Delta \rho} + 0,045 \, 34 - \frac{1,679 d}{D}} \quad (6)$$

where

γ is the approximate value of the interfacial tension;

D is the mean ring diameter;

d is the ring wire diameter;

$\Delta \rho$ is the difference in the densities of the two liquids.

NOTE The following example involves the measurement of the water/toluene system at 20 °C using a torsion balance:

At 20 °C, the densities of water and toluene are 0,998 g/cm³ and 0,867 g/cm³, respectively, resulting in a difference in density, $\Delta \rho$, of 0,131 g/cm³. The readings for W_2 and W_1 are 45,5 mN/m and 12,7 mN/m, respectively so that $\Delta W = W_2 - W_1 = 32,8$ mN/m. With an additional mass, m , of 0,5 g and a mean ring diameter, D , of 1,910 cm, A_2 is found from equation (4) to be 40,87 mN/m. When the additional mass, m , is added, the reading for A_1 is 38,2 mN/m. From equation (3), Φ is then 1,07 and, from equation (5), γ is 35,09 mN/m. Finally, if d is 0,037 0 cm, f is given by equation (6) as 1,072 6 and, consequently, γ_{zw} is 37,6 mN/m.

As is evident from this practical example, both the factor, Φ , for the apparatus and the correction factor, f , each contributes an amount of several millinewtons per metre to the value of the interfacial tension.

For the ring shown in Figure 2, the following equation (7) replaces equation (6) :

$$f = 0,725 + \sqrt{4,014 \times 10^{-4} \cdot \frac{\gamma}{\Delta\rho} + 0,01287} \quad (7)$$

Instead of calculating the correction factor, the determination of the true interfacial tension from its approximate value can be simplified using the correction values according to Table 1 [3]. This table is valid for a commercially available ring for which d is 0,037 0 cm and D is 1,909 0 cm, and is based on the Zuidema and Waters correction and, with an extension of the correction range, also on the Fox and Chrisman correction [4]. The ratio $\gamma/\Delta\rho$ is derived from the approximate value, γ , of the interfacial tension and the difference in densities, $\Delta\rho$, and the associated corrected value, $\gamma_{zw}/\Delta\rho$, is determined from the table by linear interpolation between two tabular values. Determine the true interfacial tension, γ_{zw} , by multiplying $\gamma_{zw}/\Delta\rho$ by $\Delta\rho$.

**Table 1 — Values for correction of the approximate value, γ ,
of the interfacial tension for a commercially available ring
($d = 0,037\ 0\ \text{cm}$ and $D = 1,909\ 0\ \text{cm}$)**

$\gamma/\Delta\rho$	$\gamma_{zw}/\Delta\rho$	$\gamma/\Delta\rho$	$\gamma_{zw}/\Delta\rho$
10	7,9	86	81,3
12	9,7	88	83,4
14	11,5	90	85,4
16	13,3	92	87,4
18	15,1	94	89,5
20	16,9	96	91,6
22	18,7	98	93,7
24	20,6	100	95,7
26	22,4	102	97,9
28	24,3	104	99,9
30	26,2	106	102,1
32	28,1	108	104,2
34	29,9	110	106,3
36	31,8	115	111,6
38	33,7	120	116,9
40	35,6	125	122,2
42	37,6	130	127,6
44	39,5	135	133,0
46	41,4	140	138,4
48	43,3	145	143,9
50	45,4	150	149,4
52	47,3	155	154,9
54	49,3	160	160,5
56	51,2	165	166,1
58	53,2	170	171,7
60	55,2	175	177,3
62	57,1	180	183,1
64	59,1	200	206,2
66	61,1	250	265,6
68	63,2	300	327,3
70	65,2	350	391,1
72	67,2	400	457,0
74	69,2	450	524,7
76	71,2	500	594,2
78	73,2	550	665,3
80	75,2	600	738,0
82	77,3	650	812,2
84	79,3	700	887,9

7 Precision

7.1 General

The precision of the method for determining the interfacial tension by the stirrup or ring method depends to a considerable extent on the nature of the liquid to be tested. The information given in 7.2 and 7.3 is for physically pure liquids, while that given for the reproducibility level has been obtained from interlaboratory tests using similar methods and is to be regarded as approximate.

7.2 Repeatability limit, r

The absolute difference between two independent single test results, obtained using the same method on identical test material in the same laboratory by the same operator using the same equipment within a short interval of time, will not exceed the repeatability limit, r , in more than 5 % of cases.

The repeatability limit, r , is usually less than or equal to 0,4 mN/m for the relative stirrup or ring method (see 5.2) and less than or equal to 0,2 mN/m for the absolute ring method (see 5.3).

7.3 Reproducibility limit, R

The absolute difference between two independent single test results, obtained using the same method on identical test material in different laboratories by different operators using different equipment, will not exceed the reproducibility limit, R , in more than 5 % of cases.

The reproducibility limit, R , is usually less than or equal to 2 mN/m for the relative stirrup or ring method (see 5.2) and less than or equal to 1 mN/m for the absolute ring method (see 5.3).

8 Test report

The test report shall include the following information:

- a) the type and designation of samples;
- b) a reference to this European Standard;
- c) the information on the method used;
- d) the test temperature, in degrees Celsius;
- e) the age of interface at time of measurement;
- f) the time required to reach equilibrium before measurement;
- g) the corrections used;
- h) the information if stirrup or ring was used;
- i) the individual values of interfacial tension and the arithmetic mean;
- j) any deviations from this European Standard;
- k) the date of test.

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- [5] EN 14370, *Surface active agents — Determination of surface tension.*
- [6] ISO 6889:1986, *Surface active agents and detergents — Determination of interfacial tension by drawing up liquid films.*